

Generative Conversational Recommender Systems

Julia Neidhardt

julia.neidhardt@tuwien.ac.at

January 7th, 2026



Informatics



DIGHUM
—

Why Conversational Recommendation?

- GenAI is no longer just text generation
- Modern GenAI systems:
 - interact with users
 - maintain state and memory
 - reason over goals
 - use external tools
- Conversational recommendation is a real-world setting where all of this is required

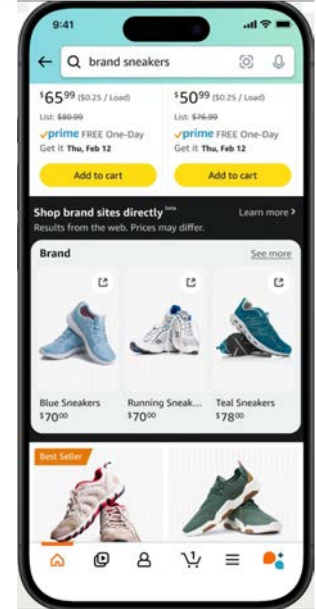
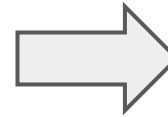
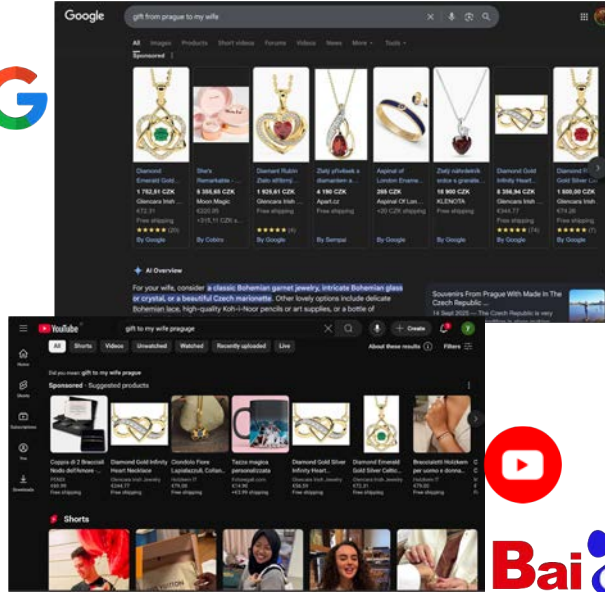
From LLMs to GenAI Systems

- A GenAI system typically includes:
 - A foundation model (LLM)
 - Memory & state tracking
 - Grounding (retrieval, databases, tools)
 - Control logic (planning, decision making)
 - Evaluation & safety mechanisms
- Why recommendation is a stress test for GenAI:
 - Goals are open-ended (“help me decide”)
 - Preferences are implicit and evolving
 - No single correct answer
 - Errors have real consequences (money, trust, fairness)
 - Interaction unfolds over multiple turns

Gen-CRS as a GenAI System



Motivating Scenario: Fragmented Search



Information
need

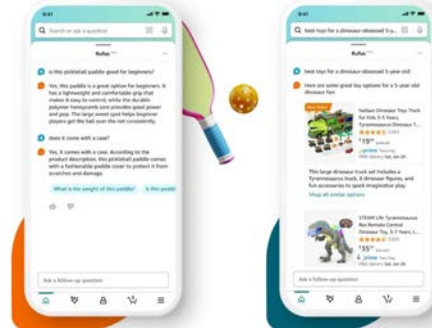
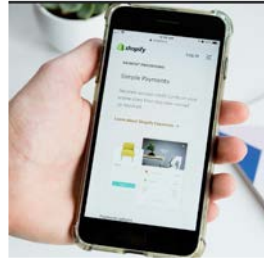
Exploration

Product Search

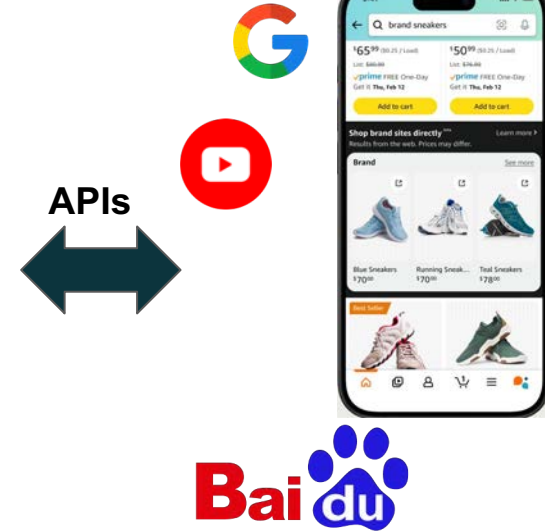
Unified Conversation + Recommendation Experience



**Information
need**



**Shopping through Rufus
conversational LLM**



What is a Conversational Recommender System (CRS)?

A Conversational Recommender System (CRS) is an interactive AI framework that engages users in **multi-turn dialogue** to **elicit - explicit preferences** and provide **personalized recommendations**.



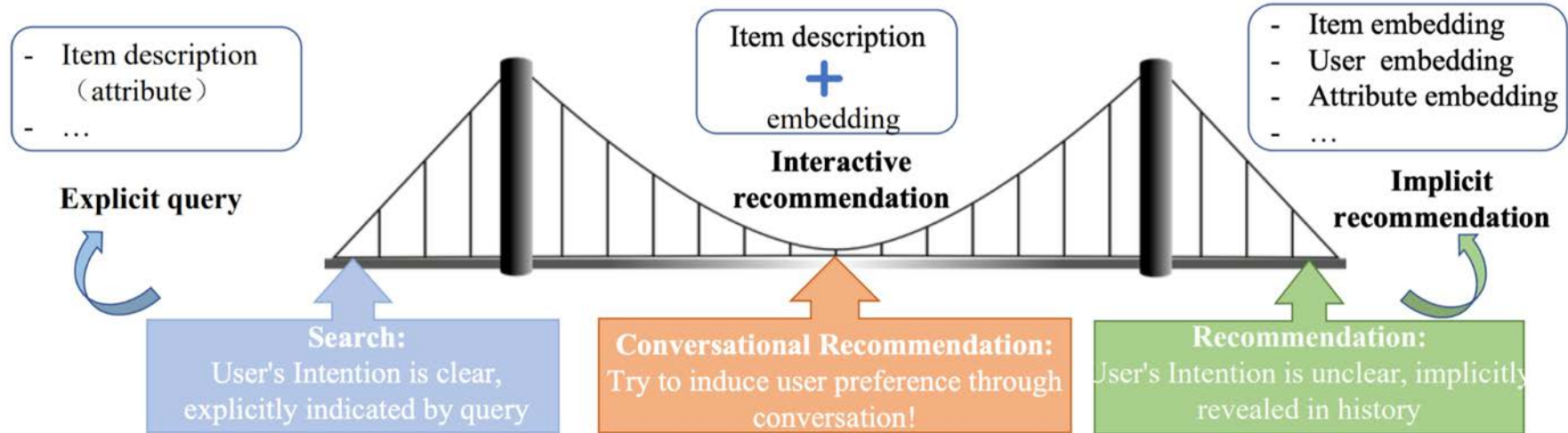
Advantages

- Assist decision-making and information-seeking
- Support product space exploration e.g., discover unexpected but relevant items
- Elicit users' nuanced or hidden preferences



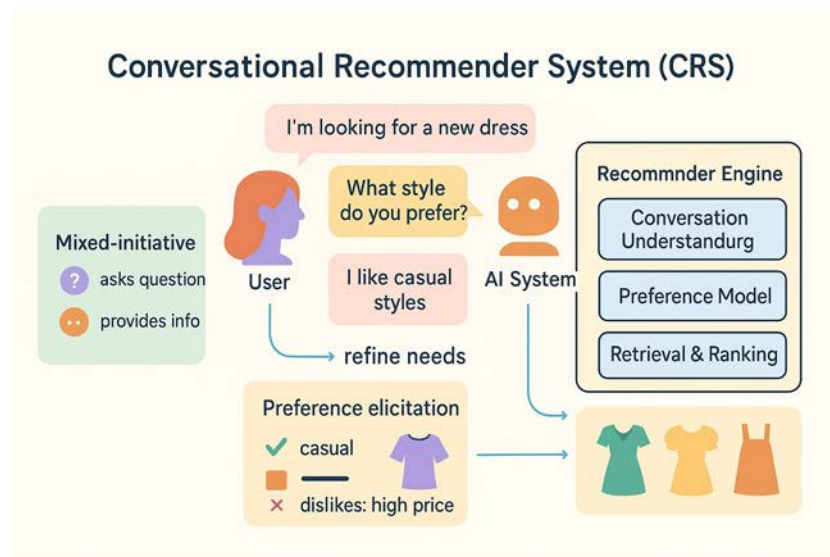
Search vs Recommendation vs Conversational Recommendation

Traditional paradigms for information-seeking:
Search (pull) or **Recommendation (push)**



Key Characteristics

- **Multi turn interaction** – conversation persists across rounds.
- **Mixed initiative** – system and user alternate turns and roles.
- **Natural language** – voice or text as the primary interface.
- **Dynamic preference elicitation** – ask, refine and adapt



CRS vs Search/QA/Chatbots



CRS (Conversational
RecSys)



CQA (Conversational Q&A)



CS (Conversational Search)



Social Chatbot



Gen-CRS

CRS vs Conversational Search

- **Common:** rank or surface relevant options in multi-turn interaction
- **Key difference:** CRS builds a user model & personalised; search focuses on query understanding & retrieval

CRS vs Conversational QA

- **Common:** dialogue to resolve information needs
- **Key difference:** QA answers factual questions; CRS elicits preferences and recommends subjective items

CRS vs Social Chatbot

- **Common:** free-form conversational exchange
- **Key difference:** CRS is goal-oriented (task success); social chat aims for open-ended chit-chat & engagement

CRS vs Gen-CRS (LLM-powered)

- **Common:** goal-oriented recommendation via dialogue
- **Key difference:** Traditional CRS uses rules/templates & separate rankers; Gen-CRS uses LLMs for free-form NLG, mixed-initiative, and tool use (with grounding to reduce hallucination)

Evolution of Conversational RS

2017-2018

Neural CRS Emergence

First neural conversational recommenders appear; separate dialogue and rec modules.

2018-2021

Evolution

Research integrates knowledge graphs & critique-based recommendation.

2022-2023

LLM Revolution

GPT-3 & ChatGPT enable generative dialogue & zero-shot recommendation.

2023-2025

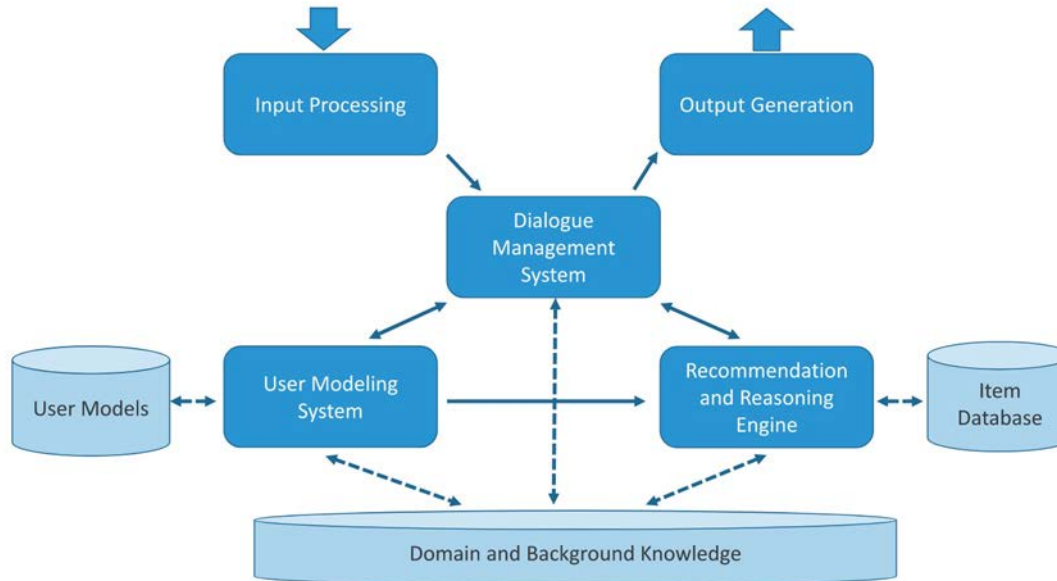
GenCRS Explosion

Unified, modular & agentic architectures flourish with LLMs & tools.



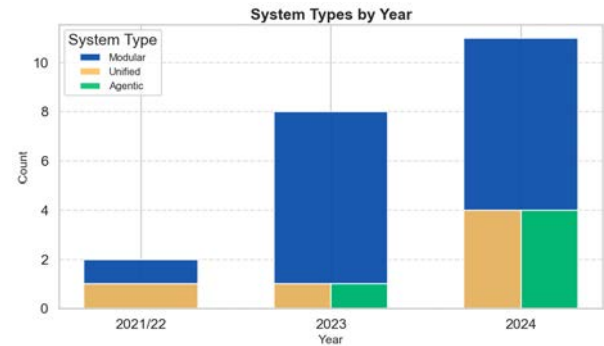
System Architecture

General CRS Architecture



System Types

- GenCRS as **emerging** topic over the past years
- **Unified:** one LLM does dialogue + recommendation end-to-end
- **Modular:** LLM handles conversation, separate recommender/retrieval handles items
- **Agentic:** LLM plans and orchestrates tools/agents (retrieval, APIs, memory)
- Most GenCRS research today is modular. Agentic GenCRS are emerging fast, while unified systems are mostly early, exploratory “LLM-as-CRS” studies.
- Most GenCRS are built as standalone prototypes, whereas agentic systems are more often embedded into existing platforms.

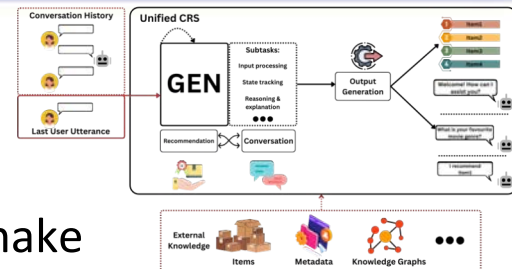


Key Questions

- Does a unified model's single LLM handle all sub-tasks end-to-end (intent detection, retrieval, ranking, NLG)?
- How well do unified systems leverage pre-trained content knowledge compared with collaborative signals?
- How interpretable and evaluable is the behavior? Do integrated architectures naturally produce richer justifications for their recommendations?

LLM as CRS

- LLM base model as zero-shot recommender
- Input: **conversation (S)**, **task (T)**, **format (F)**
- Main findings:
 - LLMs mainly rely on **content/context** knowledge to make recommendations
 - LLMs may generate **out-of-dataset** item titles, but **few hallucinated** recommendations
 - GPT-based LLMs possess **better content/context** knowledge than existing CRS
 - LLMs generally possess **weaker collaborative** knowledge than existing CRS
 - LLM recommendations suffer from **popularity bias** in CRS



Unified Gen-CRS (LLM end-to-end)

- **Pros:** simplest pipeline, fluent explanations
- **Cons:** weak grounding to real catalogues, popularity bias / misalignment, control is hard
- This motivates modular grounding (RAG/expert rankers) and later agentic planning.

System Types - Modular

Modular Separation

- **Subtasks decoupled:** user modeling, recommendation, dialogue management, response generation
- Generative models typically act as **dialogue manager, user model, or response generator**
- Recommendation modules (collaborative filtering, KG-based, neural methods) **mostly remain separate** → generative model does not replace them

System Types - Modular

Knowledge Utilization

- Collaborative: captured through **expert models, KG** etc.
- Content/context: **dialogue history embeddings, response generation by LLM**
- Clearer separation makes it easier to **optimize collaborative vs. content knowledge** sources independently
- Grounding via **RAG**

Pros & Cons

- **Pros:** interpretability, flexibility, targeted optimization, better alignment with CRS distributions
- **Cons:** pipeline complexity, coordination overhead, risk of error propagation, semantic gap between retrieval and generation

Agentic Systems

- Emerging topic in CRS research
- Modular approach that employs **specialized agents** to solve different CRS sub-tasks **orchestrated** by a central LLM agent
- Core capabilities: **planning & task decomposition**, **tool use & action execution**, **memory & state** management and **autonomy & goal-driven** behavior
- Prospect advantages:
 - **Enhanced User Experience**
 - **Adaptability & Flexibility**
 - **Contextual Precision**
 - **Explainability & Transparency**

Agentic Systems - LLM Agents Characteristics



Planning & Task Decomposition:

Break complex goals into subtasks; execute multi-step reasoning for long-horizon tasks



Memory & State Management:

Maintain context across steps; store/retrieve user preferences, domain knowledge, and feedback



Tool Use & Action Execution:

Invoke external tools/APIs; interact with real-world systems (e.g., databases, knowledge bases)



Autonomy & Goal-Driven Behavior:

Operate in closed-loop fashion; observe environment, evaluate progress, self-refine until goal completion

System Types - Agentic

Beyond Unified/Modular

- Extend with **planning & proactivity**: task decomposition, multi-step reasoning
- LLM as **controller** orchestrating expert models & external tools
- From **reactive** recommendations to **proactive, goal-driven** conversations

Additional User Goals

- **Multi-step** task completion (elicitation → recommendation → explanation → refinement)
- Continuous personalization via **memory** and **feedback**
- Extensible integration of **tools, APIs, and sub-agents**

System Types - Agentic

New Challenges

- Tool orchestration and coordination overhead
- Safety & controllability risks from autonomous behaviors
- Latency & cost from staged workflows and multi-step planning
- Full autonomy, long-term memory, and sustained planning remain challenging

Pros & Cons

- **Pros:** flexibility, proactivity, extensibility, continuous personalization
- **Cons:** complexity, safety risks, high latency/cost, limited true autonomy

Recommendation Generation

- Recommendation can be directly generated as **list**, incorporated in a system response as **fluent text** or presented **within the UI**
- Paradigms:
 - **Retrieval-based**
 - **Generative**
 - **Hybrid (retrieve -> generative re-ranking)**

Foundation Model Integration & Generative Paradigms: Key Questions

- How can foundation models be adapted to domain-specific CRS tasks?
- How much task-specific knowledge can each paradigm incorporate effectively?
- What are potential trade-offs to be considered for each paradigm?

Adaptation Paradigms

Generative Paradigms

Full fine-tuning



Prompt/Instruction tuning



In-context learning (ICL)

Generative Paradigms - Full Fine-Tuning

- All foundation model parameters are updated **end-to-end**
- Tight **semantic coupling** between recommendation and dialogue
- Applied to **smaller** or **domain-specific** models
- Pros:
 - Strong **task alignment**
 - High **adaptability** to **specific** dataset
- Cons:
 - Computationally **expensive** -> **poor scalability**
 - Risk of **bias amplification** and **catastrophic forgetting**
 - **Low generalization** capabilities and **dynamic** adaptation

Generative Paradigms - Instruction Tuning

- **Efficient** weight adaptations via supervised fine tuning on **instruction-response** pairs
- Model learns to **follow NL task instructions**
- Pros:
 - More **stable** performance on defined tasks
 - **Stronger generalization** with appropriate instructions
 - **Light-weight** updates
- Cons:
 - Requires **curated training data** and clear instructions
 - **Limited flexibility**
 - Risk of **catastrophic forgetting**

Generative Paradigms - In-Context Learning

- **No weight adaptations**/parameter updates
- Three Scenarios:
 - Zero-shot
 - One-shot
 - Few-shot
- Pros:
 - High **flexibility**
 - Fast **adaptation** & **low** cost
- Cons:
 - **Unstable** performance
 - **Sensitivity** to prompt design and examples

Knowledge and Data Foundation

Key Questions

- How does item-level knowledge improve both recommendation accuracy and response quality in generative models?
- What balance between pretrained model knowledge and external item- or user-level data is reflected in current GenCRS approaches?
- What are the key challenges in integrating structured data into generative pipelines?
- How are conversational data sets and logs consumed by generative models?

Item-Level

- Provide models with up-to-date information about item information in **dynamic** environments
- Model **collaborative signals** and make them accessible to generative models
- **Grounding & hallucination reduction**
- Model **inter-item relations** and **semantic connections** -> knowledge-aware generation
- **Item attributes**, as well as **popularity scores**
- Implicit and explicit **feedback** on items
- Widely used in **modular systems** with traditional recommendation module
- **Traditional (C)RS resources** (MovieLens, IMDb, Last.fm...)

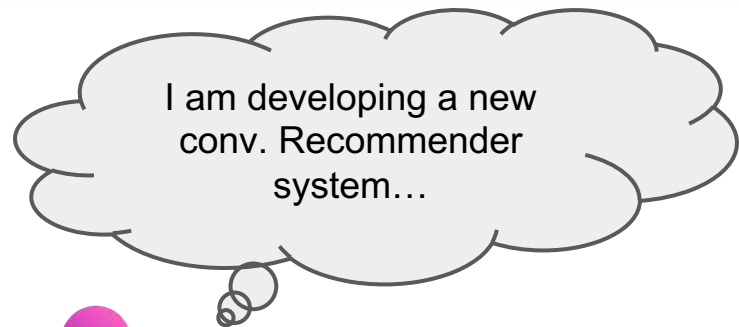
User-Level - Short- & Long-Term

- Primary focus on extracting preferences from **current conversation**
- **Static** vs. **dynamic** user profile
- **Translation** of user profiles into **NL format**
- **Building** structured user profiles **from conversation**
- **Memory management** crucial for agentic systems

User-Level - Persona & Personality

- Can be approached from a user or system perspective
- Alleviate **cold start** problems (e.g. by aligning with user demographic)
- Guide **style** or **tone** of conversation
- Enhance alignment with **user expectations**
- Risks of reproducing **stereotypes** and **biases**
- Often utilized for **user simulation**

Why Simulation? Why for Conversation?

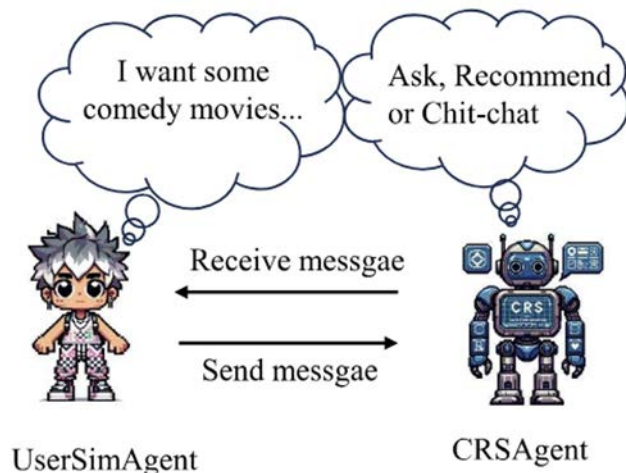


But ...

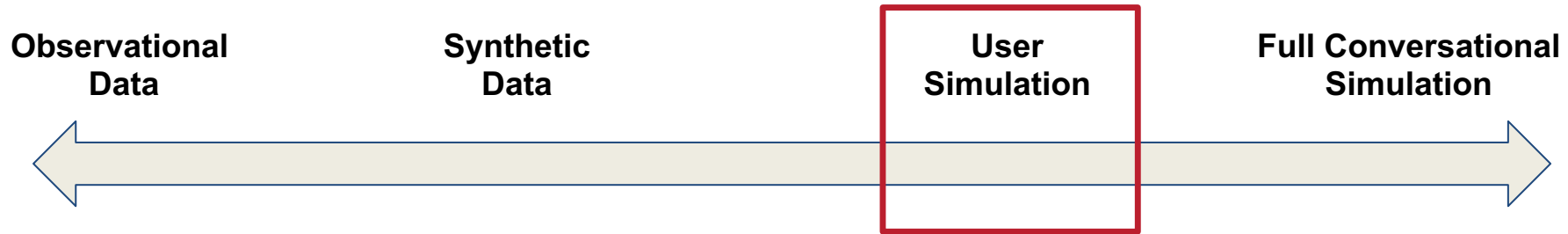
Pre GenAI: *Is there any fitting conv. dataset for evaluation?
Should I conduct an annotation study?*



With Simulation:



Towards Full Conversational Simulation



*A reactive model that **plays the role of a user** in a live conversation, producing utterances, clarifications, and preference signals in response to each system move, thereby enabling **turn-by-turn evaluation or policy learning**.*

*Less Control & Bias,
More Authentic/Real*



*More Control & Bias,
Less Authentic/Real*

Evaluation Dimensions

Output Type (What)	User-facing		Non-User-facing / hidden	
Quality Dimensions (Which Basis)	Task Effectiveness	Dialog / Task Efficiency	Conversational Quality	
	Subtask Performance	Trust & User satisfaction	Ethical Considerations & System Integrity	
Paradigms (How)	Offline	Online	User Studies & Lab Experiments	
Evaluator (Who)	Automated Metrics (e.g., NDCG, HitRate, Recall etc.,)		Humans (e.g. Fluency, Informativeness etc.,)	LLM-as-a-Judge
Stakeholders (For Whom)	Consumers	Item-Providers		Others

What to Evaluate

Output Type
(What)

User-facing

Non-User-facing / hidden

- GenCRS aims for an enjoyable, human-like, and interactive experience.
 - This requires a fundamental shift in evaluation focus.
- Evaluating the “Journey” is as important as the “Destination”
 - Journey: Dialogue coherence, interaction efficiency, user enjoyment.
 - Destination: Relevance of the final recommended items.



Evaluation Criteria

Quality Dimensions (Which Basis)

Task Effectiveness

Dialog / Task Efficiency

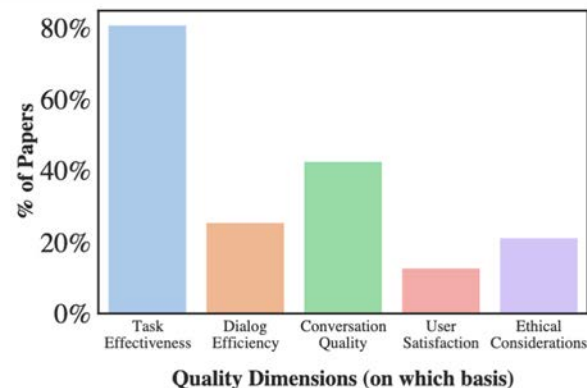
Conversational Quality

Subtask Performance

Trust & User satisfaction

Ethical Considerations
& System Integrity

- **Task Effectiveness:** *Did the user find a good item?*
- **Dialogue / Task Efficiency:** *How quickly and easily did they find it?*
- **Conversational Quality:** *Was the conversation natural and coherent?*
- **Subtask Performance:** *Did the internal modules work correctly?*
- **Trust & User Satisfaction:** *Did the user enjoy and trust the system?*
- **Ethical Considerations:** *Is the system fair, safe, and honest?*



Ethics & Integrity

Quality Dimensions (Which Basis)	Task Effectiveness	Dialog / Task Efficiency	Conversational Quality
	Subtask Performance	Trust & User satisfaction	Ethical Considerations & System Integrity

Definition: Evaluating fairness, safety, and trustworthiness, especially with powerful LLMs.

Key Areas for Evaluation:

- **Factuality & Hallucination:** *Is the system generating factually correct information? [dehbozorgi_2024, he_2023, mao_2023].*
- **Instruction Faithfulness:** *Does the model follow instructions correctly? [tsai_2024].*
- **Bias:** *Are recommendations fair and not stereotyped?*
- **User Privacy:** *Is sensitive user data protected from leakage?*
- **Robustness:** *Is the system vulnerable to adversarial manipulation?*



Status: A significant gap in current research. These critical issues are rarely evaluated rigorously and represent an imperative direction for future work.

How to Evaluate

Paradigms (How)

Offline

Online

User Studies & Lab Experiments

- **Offline Evaluation (or Simulations)**

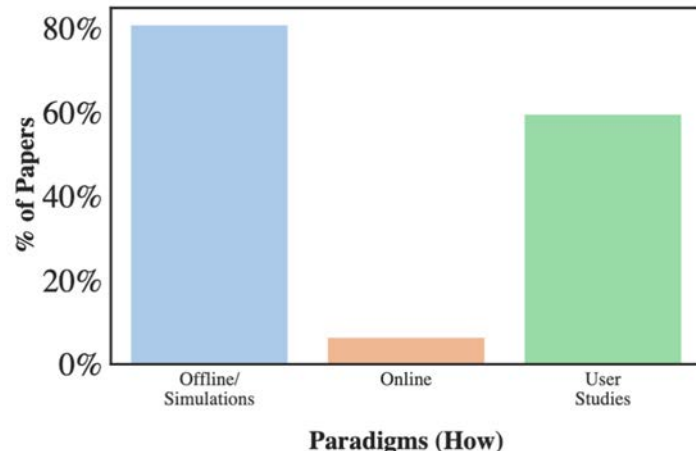
- **What:** Evaluation on static, pre-collected datasets.
- **Prevalence:** The most dominant paradigm (**~80% of papers**).

- **Online Experimentation (A/B Testing)**

- **What:** Deployment in a live environment with real users.
- **Prevalence:** The least common paradigm (**~6% of papers**).

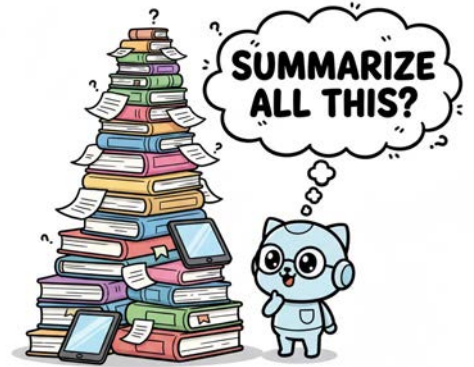
- **User Studies & Lab Experiments**

- **What:** Controlled experiments with recruited human participants.
- **Prevalence:** A crucial middle ground, used in **~60% of papers**, often to supplement offline evaluation.



Evaluation: Summary

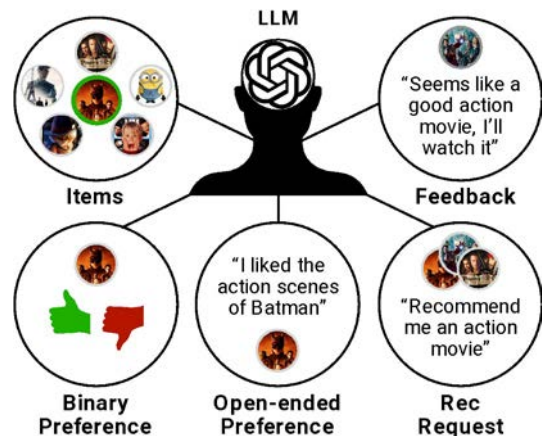
- Evaluation is evolving from a **static to a multidimensional and interactive paradigm**.
- **The Core Shift in Focus:**
 - **FROM:** Matching a single 'gold-standard' response...
 - **TO:** Measuring how helpful, human-like, and contextually appropriate a system feels.
- **The Dual Role of LLMs:**
 - LLMs now function as both the recommender agent being tested and the automated evaluator assessing performance.
- This new paradigm enables **more scalable and nuanced assessments** of the overall user experience.



Evaluation

Can we do better than held-out item prediction?

- Users may interact with conversational recommenders precisely because they struggle to articulate their preferences, or because they need to be persuaded to select a particular item;
- User studies are expensive, and generally non-reproducible
- Outside of industrial settings, user studies generally don't involve 'real' users
- User studies may be suitable for 'general knowledge' items and domains, but are unsuitable in cases where users requiring specific knowledge or expertise may be difficult to recruit



Open Challenges

- Challenges in providing diverse or novel recommendations during conversations
- Go beyond passive preference elicitation based on parsing input
- Persistent semantic gap between recommendation and response generation
- Scalable grounding methods and knowledge updates
- Lack of longitudinal benchmarks to investigate evolving preferences
- Bias, fairness and ethical considerations
- Matching system capabilities and user expecting to improve user experience and trust
- Scalability and robustness to enable feasible real-world applications
- Effective integration of multi-modal data

Summary

- Conversational recommendation represents a promising frontier in building recommender systems that are more “human-like”
- This line of research has been somewhat blown open by the excellent performance of general-purpose language models
- There’s still plenty to do (even if, arguably, less of it is about modeling...)
- Many “traditional” questions about recommender systems (evaluation, fairness, etc.) have new life in light of conversational paradigms



Link: <https://recsys-lab.at/gen-conv-recsys-tutorial>



- Based on
 - T.E. Kolb, A. Wagne, A. Banerjee, F. Nazary, J. Neidhardt, Y. Deldjoo, and T. Di Noia. 2025. **A Tutorial on Recent Advances in Generative Conversational Recommender Systems**. In *Proceedings of the Nineteenth ACM Conference on Recommender Systems (RecSys '25)*. ACM, New York, NY, USA, 1420–1422. <https://doi.org/10.1145/3705328.3748010>

Background

Upcoming survey paper:

Advancements in Conversational Recommender Systems Using Generative Models: A Systematic Literature Review

AHMADOU WAGNE, TU Wien, Austria

THOMAS ELMAR KOLB, TU Wien, Austria

ASHMI BANERJEE, Technical University of Munich, Germany

FATEMEH NAZARY, Polytechnic University of Bari, Italy

JULIA NEIDHARDT, TU Wien, Austria

YASHAR DELDJOO, Polytechnic University of Bari, Italy

Link: <https://recsys-lab.at/gen-conv-recsys-tutorial>

